13.6 Choosing a procedure

None of the procedures of this chapter is best in general, because they answer different questions. This section compares them on one data set, collects the comparison into a guide, and explains why several procedures common in software are not developed here.

13.6.1 Decide the family first

Every guarantee in this chapter is for a *family* of inferences, and is only as good as the honesty with which the family is defined. A short list of functions written down before the data arrive is a **planned** family. All $g(g-1)/2$ pairwise differences form a family with its own exact method. A function chosen *because* the data make it look interesting belongs to the family of everything the analyst might have chosen, usually a whole subspace; this is **post hoc** inference, or data snooping. And a long list screened for leads worth following up calls for a different error rate altogether.

> *Warning.* The family must be fixed before the data are examined, and reported. Choosing the adjustment after seeing the results, for example reporting Bonferroni's intervals for the three comparisons that turned out largest, reproduces the problem that adjustment was meant to solve. So does splitting one analysis into several "families" so that each is small.

13.6.2 One data set, five sets of intervals

The education example of Example (Education and the perceived position of a candidate) and Example (All pairs of education levels) makes the differences concrete. The table gives, for each method, the multiplier of the standard error for all 21 pairwise differences, the resulting half-width for two pairs with very different precision, and the number of pairs declared different at familywise level 0.05 (false discovery rate 0.05 for BH). The pair of levels 1 and 2 involves the two smallest groups, with standard error 0.420; the pair of levels 3 and 6 involves two large groups, with standard error 0.124.

method	multiplier	ratio to t	half-width, 1 vs 2	half-width, 3 vs 6	pairs declared different
unadjusted t	1.962	1.00	0.824	0.244	14
Tukey–Kramer	2.955	1.51	1.241	0.368	9
Šidák	3.039	1.55	1.276	0.378	8
Bonferroni	3.046	1.55	1.279	0.379	8
Scheffé	3.557	1.81	1.494	0.442	6
Holm					10
Benjamini–Hochberg					13

The ordering is the one the theory predicts: the method designed for pairs is shortest, the general finite-family methods are a few percent longer, and the method covering every contrast is much

longer. Holm's procedure rejects more by stepping down, and BH by controlling a weaker error rate. The Šidák row relies on Šidák's inequality for correlated normal variables, stated without proof in Section 13.4.

```
import itertools
import numpy as np
import statsmodels.api as sm
from scipy import stats
anes = sm.datasets.anes96.load_pandas().data
y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1       # 0, ..., 6
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g                                            # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu             # pooled variance estimate
s = np.sqrt(s2)
alpha = 0.05
c_sch = np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu))   # Scheffe multiplier,
     q = g - 1
pairs = list(itertools.combinations(range(g), 2))
m = len(pairs)                                        # 21 comparisons
diff = np.array([means[k] - means[l] for k, l in pairs])
se_d = np.array([s * np.sqrt(1 / n_k[k] + 1 / n_k[l]) for k, l in pairs])
t = diff / se_d
mult = {                                              # half-width = multiplier x se
    "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
    "Bonferroni": stats.t.ppf(1 - alpha / (2 * m), nu),
    "Sidak": stats.t.ppf((1 + (1 - alpha) ** (1 / m)) / 2, nu),
    "Tukey-Kramer": stats.studentized_range.ppf(1 - alpha, g, nu) / np.sqrt(2),
    "Scheffe": c_sch,
}
for name, c in mult.items():
    print(f"{name:13s} multiplier {c:.3f}   pairs declared different: {np.sum(np.abs
    (t) > c)}")
```

Run this code online

13.6.3 A decision guide

The choice can be put as a short sequence of questions.

(1) *Is there one question?* Then use the ordinary t or F procedure of Chapter 11 or Chapter 12. No adjustment is needed.

(2) *Will the functions be chosen after seeing the data?* If they lie in a known subspace, use Scheffé's intervals for that subspace, choosing the smallest subspace that contains every function one might pick (Section 13.2).

(3) *Is the family all pairs of means?* Use Tukey's intervals if the design is balanced and the means are independent, Tukey–Kramer if the one-way layout is unbalanced, and the exact multivariate t computation otherwise.

(4) *Is the family a short list fixed in advance?* Use Bonferroni's (or Šidák's) intervals, or Holm's tests if intervals are not needed, or the exact multivariate t value if software is at

hand.

(5) *Is the family long, and the goal a list of leads for further study?* Control the false discovery rate with the Benjamini–Hochberg procedure, or with the Benjamini–Yekutieli version if dependence among the tests is strong and of unknown form.

Where intervals are available they are preferable to tests alone, because they report the size of each difference as well as its sign (Proposition 13.1.2). Adjusted p-values, Eq. 13.4.3 and Eq. 13.5.2, let readers apply a different level. And a report should say how many comparisons were made in all, not only which were significant.

13.6.4 Planned versus post hoc

A post hoc comparison is chosen because it looks large, and its t statistic has the distribution of a maximum. In the education example the most extreme contrast has $t = 6.88$. Under equal means its square is not an $F(1, 937)$ variable but 6 times an $F(6, 937)$ variable (Corollary 13.2.3), and treating it as planned would overstate the evidence by exactly the amount Scheffé's multiplier corrects. The same happens at a larger scale when regressors are selected by their t statistics; Section 29.5 takes up inference after model selection.

13.6.5 Procedures not developed in this book

Several procedures widely available in software are deliberately left out. They control the familywise error rate only weakly, or not at all.

Fisher's protected LSD controls the familywise error rate only weakly, as Example (The protected LSD without strong control) and Exercise (C1) show, except with three groups (Exercise (B1)).

The Newman–Keuls procedure orders the means and tests the range of each set of w adjacent means against $q_\alpha(w, \nu)\, s/\sqrt{m}$, stepping down from $w = g$ and never testing inside a set whose range was not significant. Under the global null it rejects exactly when Tukey's test does, so its control is weak; with means in well-separated groups, each small group is tested nearly unprotected.

Example. (Newman–Keuls without strong control)

Take $g = 10$ groups of 5 observations, with the true means equal in pairs and the five pairs far apart, and $\nu = 40$. The larger ranges are always significant, so the procedure reaches every pair and tests it with $q_\alpha(2, \nu)$, the critical value of a single unadjusted comparison. In 10000 simulated data sets, the Newman–Keuls procedure declares some pair of equal means different in a proportion 0.219 of them. Five independent level-0.05 tests would err with probability 0.226; the shared s lowers the rate slightly, as Proposition 13.4.3 implies. Tukey's intervals err in 0.011 of the data sets.

```
import numpy as np
from scipy import stats

alpha, g, m_rep, reps = 0.05, 10, 5, 10_000
nu = g * (m_rep - 1)
mu = np.repeat(np.arange(5) * 20.0, 2)                # pairs 0,0, 20,20, 40,40, ...
```

```
def newman_keuls(means, s, m_rep, q_crit):
    """Pairs (k, l) declared different by the Newman-Keuls step-down procedure;
    q_crit[w] is the upper-alpha point of the studentized range of w means."""
    g = len(means)
    order = np.argsort(means)
    ybar = means[order]
    se = s / np.sqrt(m_rep)
    kept = []                                          # windows found homogeneous
    different = set()
    for width in range(g, 1, -1):                      # ranges of width, width-1,
    ..., 2
        crit = q_crit[width] * se
        for i in range(g - width + 1):
            j = i + width - 1
            if any(a <= i and j <= b for a, b in kept):
                continue                               # inside a homogeneous window
            if ybar[j] - ybar[i] > crit:
                different.add((min(order[i], order[j]), max(order[i], order[j])))
            else:
                kept.append((i, j))
    return different

q_crit = {w: stats.studentized_range.ppf(1 - alpha, w, nu) for w in range(2, g + 1)}
rng = np.random.default_rng(20260919)
q_tukey = stats.studentized_range.ppf(1 - alpha, g, nu)
nk_false = tukey_false = 0
for r in range(reps):
    y = mu[:, None] + rng.normal(size=(g, m_rep))
    means = y.mean(axis=1)
    s = np.sqrt(np.sum((y - means[:, None]) ** 2) / nu)
    true_pairs = [(2 * i, 2 * i + 1) for i in range(5)]
    nk = newman_keuls(means, s, m_rep, q_crit)
    nk_false += any(pair in nk for pair in true_pairs)
    tukey_false += any(abs(means[a] - means[b]) > q_tukey * s / np.sqrt(m_rep)
                       for a, b in true_pairs)
print(f"P(some false difference): Newman-Keuls {nk_false / reps:.3f}, "
      f"Tukey {tukey_false / reps:.3f}")
```

Run this code online

With $2J$ means in J separated pairs, the familywise error rate of Newman–Keuls approaches $1-(1-\alpha)^J$ as the degrees of freedom grow (Exercise (B2)). No bound below one holds uniformly in g.

Duncan's multiple range test. Duncan's procedure has the structure of Newman–Keuls but raises the level used for a set of w means to $1-(1-\alpha)^{w-1}$. Its first step, the range of all g means, is therefore a test at level $1-(1-\alpha)^{g-1}$, and if it rejects, at least the most extreme pair is declared different. Under the global null its familywise error rate is exactly $1-(1-\alpha)^{g-1}$, which is 0.370 for $g=10$ and $\alpha=0.05$. It does not control the rate even weakly.

These procedures buy power over Tukey's by relaxing the error rate when some hypotheses are false. If that trade is wanted, it is better made openly, by controlling the false discovery rate, whose meaning is clear. Closed testing (Theorem 13.4.4) with range or F local tests gives step-down procedures that are more powerful than Tukey's and still control the familywise error rate strongly; the books cited in the notes describe them.

A Bayesian response to multiplicity is a hierarchical model that treats the group means as draws from a common distribution and shrinks the estimated differences toward each other (Exercise (C1)). Chapter 32 develops random-effects models of this kind (Proposition 32.6.1), and the conjugate analysis of Theorem 7.6.2 is their simplest case.

13.6.6 Exercises

13.6.6.1 A. Check your understanding

Exercise A1.

Choose a procedure for each situation, and say what family its guarantee refers to: *(a)* a trial with three doses and a placebo, where the protocol specifies the three comparisons with placebo; *(b)* eight fertilizers compared in a balanced field trial, with interest in every pair; *(c)* 5000 genes screened for association with a trait, to select candidates for a second study; *(d)* a contrast suggested by a plot of the group means.

13.6.6.2 B. Practice

Exercise B1.

Show that under the global null the familywise error rate of Duncan's procedure is $1-(1-\alpha)^{g-1}$ in a balanced one-way layout.

Exercise B2.

In Example (Newman–Keuls without strong control) with J pairs, let the separation between pairs tend to infinity. Show that the familywise error rate of Newman–Keuls converges to $1-\Pr\bigl(|T_j| \le q_\alpha(2,\nu)/\sqrt{2},\ j=1,\dots,J\bigr)$, where the T_j are t statistics with independent numerators and a common s. Using Proposition 13.4.3, show that the limit is at most $1-(1-\alpha)^J$, and that it tends to this value as $\nu \to \infty$.

13.6.6.3 C. Going deeper

Exercise C1.

In the balanced one-way layout with known σ^2, give the group means the prior $\mu_k \overset{\text{iid}}{\sim} \mathrm{N}(\mu_0, \tau^2)$, with μ_0 and τ^2 known. Find the posterior distribution of $\mu_k - \mu_l$ and compare the width and centre of its central 95% posterior interval with the unadjusted confidence interval. Explain in what sense the shrinkage addresses the multiplicity of comparisons, and in what sense it does not.